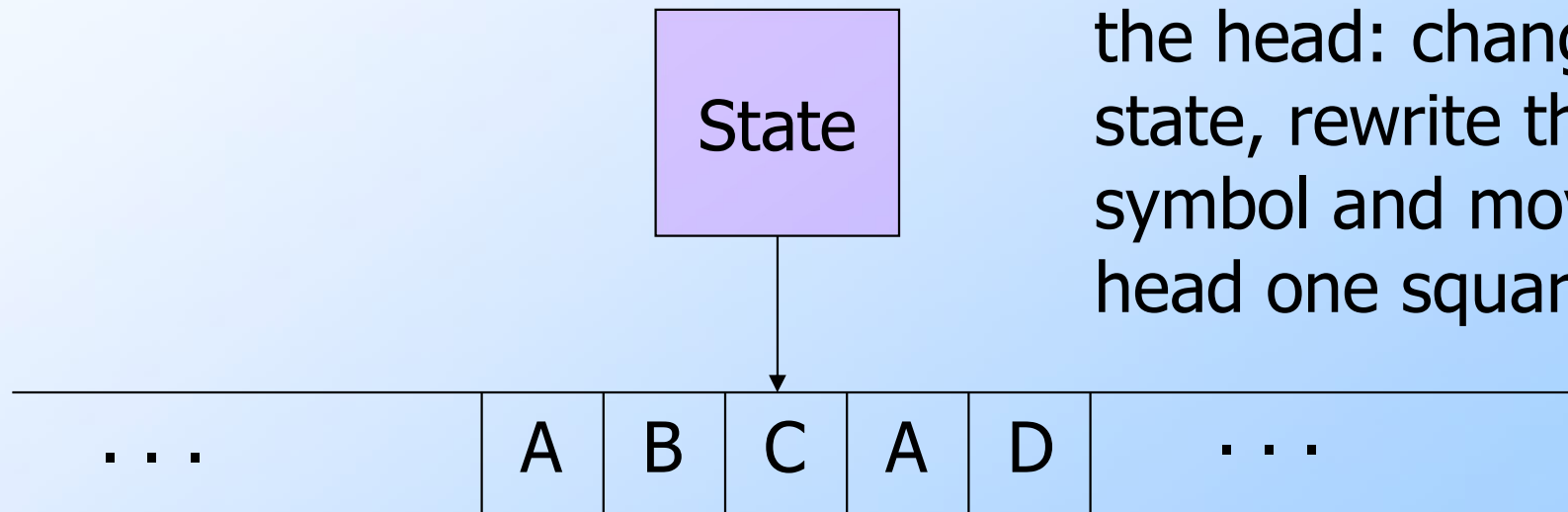


Turing-Machine Theory

- The purpose of the theory of Turing machines is to prove that certain specific languages have no algorithm.
- Start with a language about Turing machines themselves.
- Reductions are used to prove more common questions undecidable.

Picture of a Turing Machine

Action: based on the state and the tape symbol under the head: change state, rewrite the symbol and move the head one square.



Infinite tape with squares containing tape symbols chosen from a finite alphabet

Why Turing Machines?

- Why not deal with C programs or something like that?
- **Answer:** You can, but it is easier to prove things about TM's, because they are so simple.
 - And yet they are as powerful as any computer.
 - More so, in fact, since they have infinite memory.

Then Why Not Finite-State Machines to Model Computers?

- In principle, you could, but it is not instructive.
- Programming models don't build in a limit on memory.
- In practice, you can go to Fry's and buy another disk.
- But finite automata vital at the chip level (model-checking).

Turing-Machine Formalism

- A TM is described by:
 1. A finite set of *states* (Q , typically).
 2. An *input alphabet* (Σ , typically).
 3. A *tape alphabet* (Γ , typically; contains Σ).
 4. A *transition function* (δ , typically).
 5. A *start state* (q_0 , in Q , typically).
 6. A *blank symbol* (B , in $\Gamma - \Sigma$, typically).
 - All tape except for the input is blank initially.
 7. A set of *final states* ($F \subseteq Q$, typically).

Conventions

- $a, b, \dots$ are input symbols.
- $\dots, X, Y, Z$ are tape symbols.
- $\dots, w, x, y, z$ are strings of input symbols.
- $\alpha, \beta, \dots$ are strings of tape symbols.

The Transition Function

- Takes two arguments:
 1. A state, in Q .
 2. A tape symbol in Γ .
- $\delta(q, Z)$ is either undefined or a triple of the form (p, Y, D) .
 - p is a state.
 - Y is the new tape symbol.
 - D is a *direction*, L or R.

Actions of the PDA

- If $\delta(q, Z) = (p, Y, D)$ then, in state q , scanning Z under its tape head, the TM:
 1. Changes the state to p .
 2. Replaces Z by Y on the tape.
 3. Moves the head one square in direction D .
 - $D = L$: move left; $D = R$: move right.

Example: Turing Machine

- This TM scans its input right, looking for a 1.
- If it finds one, it changes it to a 0, goes to final state f , and halts.
- If it reaches a blank, it changes it to a 1 and moves left.

Example: Turing Machine – (2)

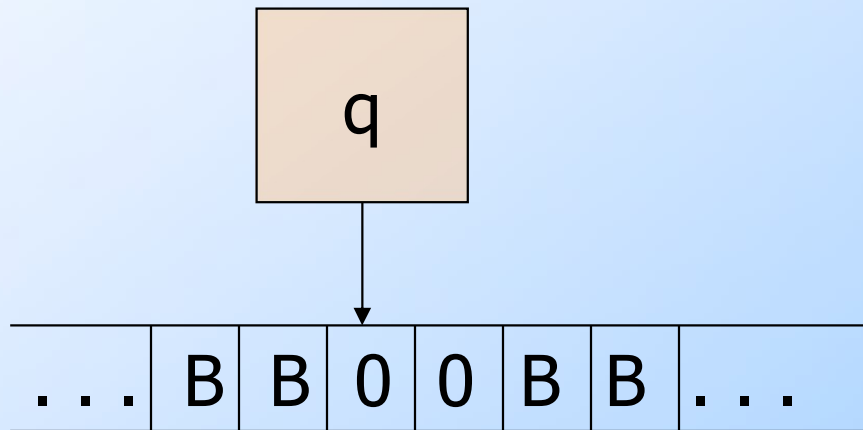
- States = $\{q \text{ (start)}, f \text{ (final)}\}$.
- Input symbols = $\{0, 1\}$.
- Tape symbols = $\{0, 1, B\}$.
- $\delta(q, 0) = (q, 0, R)$.
- $\delta(q, 1) = (f, 0, R)$.
- $\delta(q, B) = (q, 1, L)$.

Simulation of TM

$$\delta(q, 0) = (q, 0, R)$$

$$\delta(q, 1) = (f, 0, R)$$

$$\delta(q, B) = (q, 1, L)$$

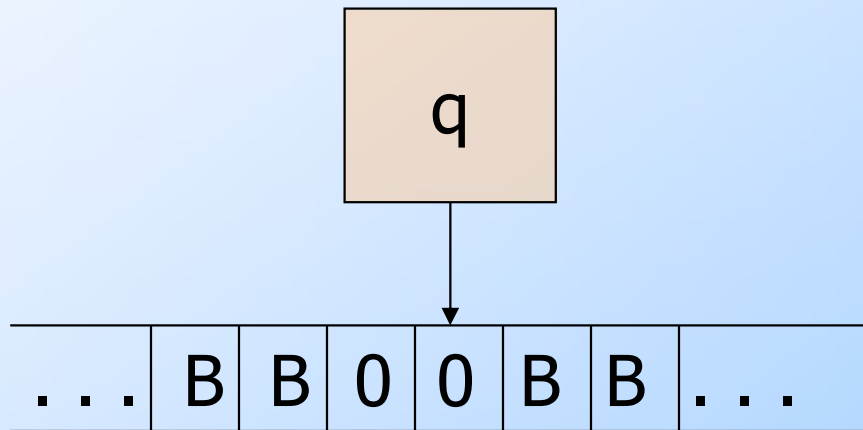


Simulation of TM

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$$\delta(q, 1) = (f, 0, R)$$

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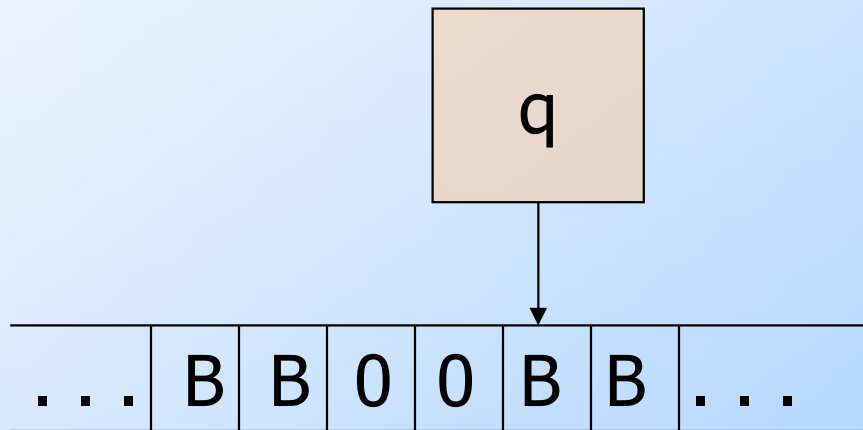


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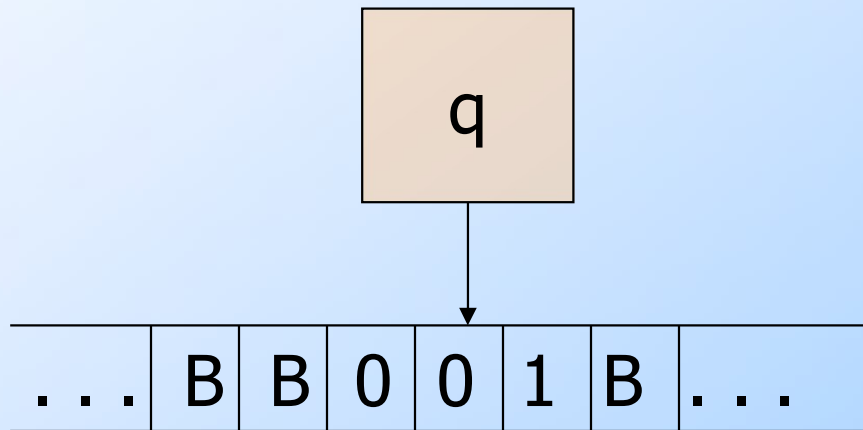


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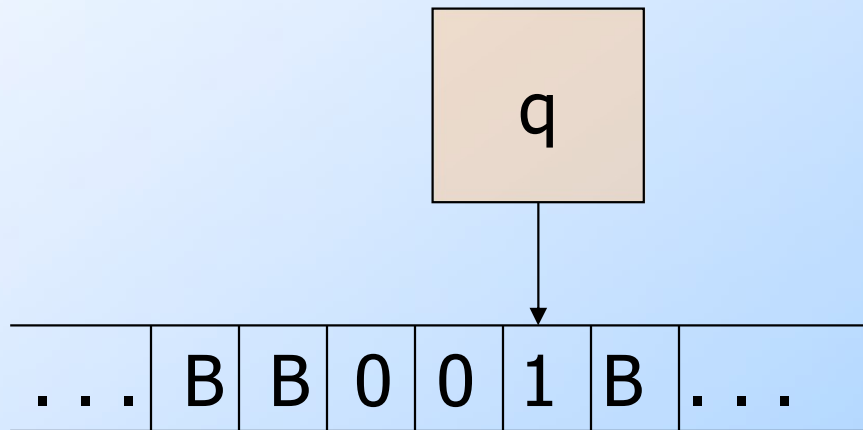


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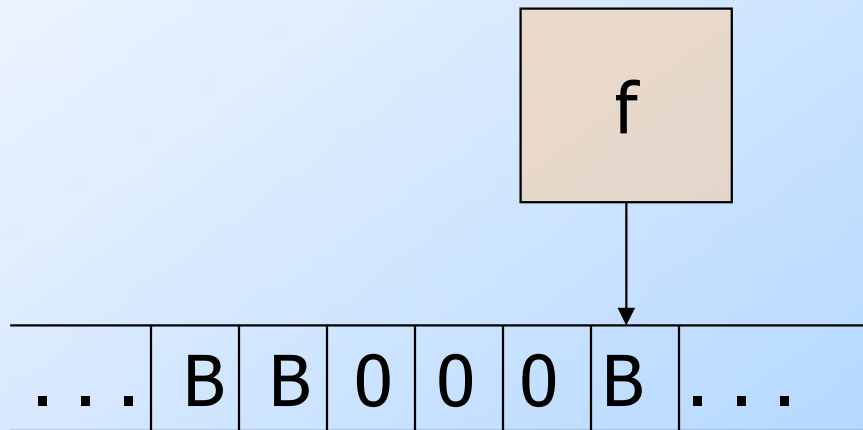


Simulation of TM

$$\delta(q, 0) = (q, 0, R)$$

$$\delta(q, 1) = (f, 0, R)$$

$$\delta(q, B) = (q, 1, L)$$



No move is possible.
The TM halts and
accepts.

Instantaneous Descriptions of a Turing Machine

- Initially, a TM has a tape consisting of a string of input symbols surrounded by an infinity of blanks in both directions.
- The TM is in the start state, and the head is at the leftmost input symbol.

TM ID's – (2)

- An ID is a string $\alpha q \beta$, where $\alpha \beta$ is the tape between the leftmost and rightmost nonblanks (inclusive).
- The state q is immediately to the left of the tape symbol scanned.
- If q is at the right end, it is scanning B .
 - If q is scanning a B at the left end, then consecutive B 's at and to the right of q are part of α .

TM ID's – (3)

- As for PDA's we may use symbols $\vdash$ and $\vdash^*$ to represent “becomes in one move” and “becomes in zero or more moves,” respectively, on ID's.
- **Example:** The moves of the previous TM are $q00 \vdash 0q0 \vdash 00q \vdash 0q01 \vdash 00q1 \vdash 000f$

Formal Definition of Moves

1. If $\delta(q, Z) = (p, Y, R)$, then
 - $\alpha q Z \beta \vdash \alpha Y p \beta$
 - If Z is the blank B , then also $\alpha q \vdash \alpha Y p$
2. If $\delta(q, Z) = (p, Y, L)$, then
 - For any X , $\alpha X q Z \beta \vdash \alpha p X Y \beta$
 - In addition, $q Z \beta \vdash p B Y \beta$